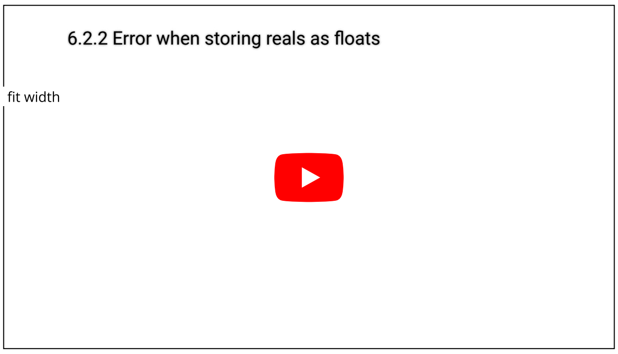
6.2.2 Error in storing a real number as a floating point number



Remark 6.2.2.1. We consider the case where a real number is truncated to become the stored floating point number. This makes the discussion a bit simpler.

Let positive χ be represented by

$$\chi = .\delta_0\delta_1\cdots \times 2^e,$$

where δ_i are binary digits and $\delta_0 = 1$ (the mantissa is normalized). If t binary digits are stored by our floating point system, then

$$\check{\chi} = .\delta_0\delta_1\cdots\delta_{t-1} \times 2^e$$

is stored (if truncation is employed). If we let $\delta\chi = \chi - \check{\chi}$. Then

$$\begin{array}{rcl} \delta\chi & = & \underbrace{.\delta_0\delta_1\cdots\delta_{t-1}\delta_t\cdots \times 2^e}_{\chi} - \underbrace{.\delta_0\delta_1\cdots\delta_{t-1} \times 2^e}_{\check{\chi}} \\ & = & \underbrace{.0\cdots 00}_{t}\delta_t\cdots \times 2^e \\ & < & \underbrace{.0\cdots 01}_{t} \times 2^e = 2^{-t}2^e. \end{array}$$

Since χ is positive and $\delta_0 = 1$,

$$\chi = .\delta_0\delta_1\cdots \times 2^e \geq \frac{1}{2} \times 2^e.$$

Thus,

$$\frac{\delta\chi}{\chi} \leq \frac{2^{-t}2^e}{\frac{1}{2}2^e} = 2^{-(t-1)},$$

which can also be written as

$$\delta\chi \leq 2^{-(t-1)}\chi.$$

A careful analysis of what happens when χ equals zero or is negative yields

$$|\delta\chi| \leq 2^{-(t-1)}|\chi|.$$

Example 6.2.2.2. The number $4/3 = 1.3333\cdots$ can be written as

$$\begin{array}{l} 1.3333\cdots \\ \quad = \\ 1 + \frac{0}{2} + \frac{1}{4} + \frac{0}{8} + \frac{1}{16} + \cdots \\ \quad = \quad < \text{convert to binary representation} > \\ 1.0101\cdots \times 2^0 \\ \quad = \quad < \text{normalize} > \\ .10101\cdots \times 2^1 \end{array}$$

Now, if $t = 4$ then this would be truncated to

$$.1010 \times 2^1,$$

which equals the number

$$\begin{array}{l} .101 \times 2^1 \quad = \\ \frac{1}{2} + \frac{0}{4} + \frac{1}{8} + \frac{0}{16} \times 2^1 \\ \quad = \\ 0.625 \times 2 \quad = \quad < \text{convert to decimal} > \\ 1.25 \end{array}$$

The relative error equals

$$\frac{1.333\cdots - 1.25}{1.333\cdots} = 0.0625.$$

If $\check{\chi}$ is computed by rounding instead of truncating, then

$$|\delta\chi| \leq 2^{-t}|\chi|.$$

We can abstract away from the details of the base that is chosen and whether rounding or truncation is used by stating that storing χ as the floating point number $\check{\chi}$ obeys

$$|\delta\chi| \leq \epsilon_{\rm mach}|\chi|$$

where $\epsilon_{\rm mach}$ is known as the *machine epsilon* or *unit roundoff*. When single precision floating point numbers are used $\epsilon_{\rm mach} \approx 10^{-8}$, yielding roughly eight decimal digits of accuracy in the stored value. When double precision floating point numbers are used $\epsilon_{\rm mach} \approx 10^{-16}$, yielding roughly sixteen decimal digits of accuracy in the stored value.

Example 6.2.2.3. The number $4/3 = 1.3333\cdots$ can be written as

$$\begin{array}{l} 1.3333\cdots \\ = \\ 1 + \frac{0}{2} + \frac{1}{4} + \frac{0}{8} + \frac{1}{16} + \cdots \\ = \quad < \text{convert to binary representation} > \\ 1.0101\cdots \times 2^0 \\ = \quad < \text{normalize} > \\ .10101\cdots \times 2^1 \end{array}$$

Now, if $t = 4$ then this would be rounded to

$$.1011 \times 2^1,$$

which is equals the number

$$\begin{array}{l} .1011 \times 2^1 \quad = \\ \frac{1}{2} + \frac{0}{4} + \frac{1}{8} + \frac{1}{16} \times 2^1 \\ = \\ 0.6875 \times 2 \quad = \quad < \text{convert to decimal} > \\ 1.375 \end{array}$$

The relative error equals

$$\frac{|1.333\cdots - 1.375|}{1.333\cdots} = 0.03125.$$

Definition 6.2.2.4. Machine epsilon (unit roundoff). The machine epsilon (unit roundoff), $\epsilon_{\rm mach}$, is defined as the smallest positive floating point number χ such that the floating point number that represents $1 + \chi$ is greater than one.

Remark 6.2.2.5. The quantity $\epsilon_{\rm mach}$ is machine dependent. It is a function of the parameters characterizing how a specific architecture converts reals to floating point numbers.

Homework 6.2.2.1. Assume a floating point number system with $\beta = 2$, a mantissa with t digits, and truncation when storing.

- Write the number 1 as a floating point number in this system.
- What is the $\epsilon_{\rm mach}$ for this system?

▼ Solution

- Write the number 1 as a floating point number.

  Answer:

  $$\underbrace{.10\cdots0}_{\begin{array}{c} t \\ \text{digits} \end{array}} \times 2^1.$$

- What is the $\epsilon_{\rm mach}$ for this system?

  Answer:

  $$\underbrace{\underbrace{.10\cdots0}_{t \text{ digits}} \times 2^1}_{1} + \underbrace{\underbrace{.00\cdots1}_{t \text{ digits}} \times 2^1}_{2^{-(t-1)}} = \underbrace{\underbrace{.10\cdots1}_{t \text{ digits}} \times 2^1}_{> 1}$$

  and

  $$\underbrace{\underbrace{.10\cdots0}_{t \text{ digits}} \times 2^1}_{1} + \underbrace{\underbrace{.00\cdots0}_{t \text{ digits}}11\cdots \times 2^1}_{< 2^{-(t-1)}} = \underbrace{\underbrace{.10\cdots0}_{t \text{ digits}}11\cdots \times 2^1}_{\text{truncates to } 1}$$

  Notice that

  $$\underbrace{.00\cdots1}_{t \text{ digits}} \times 2^1$$

  can be represented as

  $$\underbrace{.10\cdots0}_{t \text{ digits}} \times 2^{-(t-2)}$$

  and

$$\underbrace{.00\cdots0}_{t\text{ digits}}11\cdots\times 2^1$$

as

$$\underbrace{.11\cdots1}_{t\text{ digits}}\times 2^{-(t-1)}$$

Hence $\epsilon_{\mathbf{mach}} = 2^{-(t-1)}$.